



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

14-06-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	B	C	C	A	A	A	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	A	A	D	B	C	B	B	D

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	C	D	A	B	C	A	D	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	D	A	B	B	C	C	B	A

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	B	C	C	B	D	B	A	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	A	C	D	B	D	A	C	A

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (C)**

$$\frac{T_{\max.}}{2} - mg = m(2) \quad \dots(i)$$

$$T_{\max.} - mg = ma \quad \dots(ii)$$

From (i), we get, $T_{\max} = 2m(g + 2)$

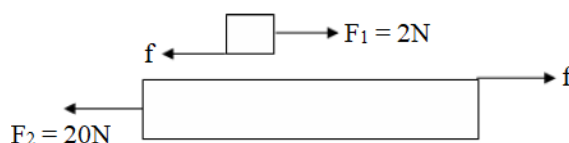
putting above value of $T_{\max.}$ in equation (ii) we get,

$$2m(g + 2) - mg = ma$$

$$\text{i.e. } a = 2g + 4 - g = g + 4 = 13.8 \text{ m/s}^2$$

2. **Ans (B)**

Free body diagrams of the two bodies are as follows :



Let acceleration of both the blocks towards left be a . Then

$$a = \frac{f - 2}{2} = \frac{20 - f}{4}$$

$$\text{or } 2f - 4 = 20 - f$$

$$\text{or } 3f = 24 \text{ or } f = 8\text{N}$$

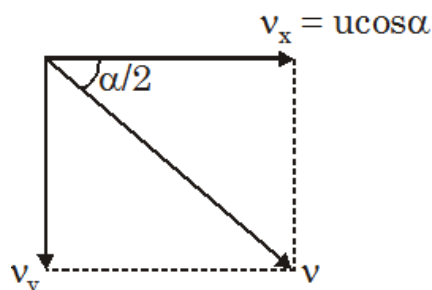
Maximum friction between two blocks can be :

$$f_{\max} = \mu mg = 0.5 \times 2 \times 10 = 10 \text{ N}$$

Therefore, friction force between the two blocks is 8 N.

3. **Ans (C)**

Horizontal component of velocity i.e. $u \cos \alpha$ remains unchanged. At highest point vertical component of velocity is zero. Let v_y be the vertical component of velocity at the desired instant.



$$\frac{v_y}{v_x} = \tan \frac{\alpha}{2}$$

$$\therefore v_y = v_x \tan \frac{\alpha}{2} = u \cos \alpha \tan \frac{\alpha}{2}$$

Now increase in kinetic energy = work done by gravity

$$\therefore \text{The desired work done } w = \frac{1}{2} m v_y^2$$

4. **Ans (B)**

From conservation of mechanical energy

$$\frac{1}{2} k r^2 = \frac{1}{4\pi\epsilon_0} \left[\frac{q^2}{r} - \frac{q^2}{r+r} \right]$$

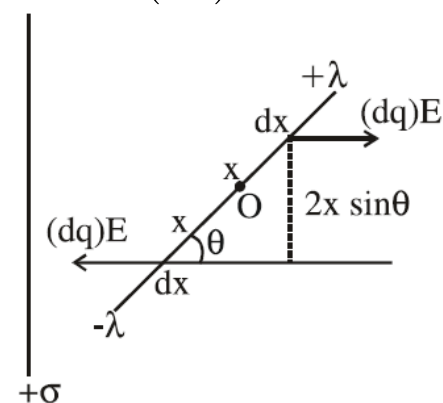
$$\text{or } \frac{1}{2} k r^2 = \frac{q^2}{8\pi\epsilon_0 r}$$

$$K = \frac{q^2}{4\pi\epsilon_0 r^3}$$

5. **Ans (C)**

$$dx = (dq) E (2x \sin \theta)$$

$$= (\lambda \cdot dx) \left(\frac{\sigma}{2\epsilon_0} \right) (2x \sin \theta)$$



$$= \frac{\sigma \lambda x \sin \theta}{\epsilon_0} \cdot dx \therefore \tau = \int_0^l d\tau = \frac{\sigma \lambda^2 \sin \theta}{2\epsilon_0}$$

6. **Ans (C)**

If resistance of ammeter is r then

$$20 = (R + r) 4$$

$$\Rightarrow R + r = 5 \Rightarrow R < 5 \Omega$$

7. **Ans (A)**

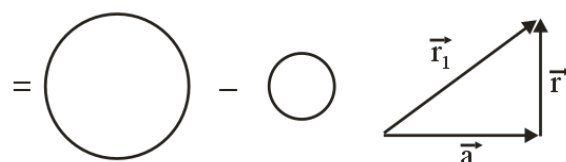
The equation of pressure variation due to sound is

$$p = -B \frac{\partial s}{\partial x} = -B \frac{\partial}{\partial x} [s_0 \sin^2(\omega t - kx)]$$

$$= B k s_0 \sin(2\omega t - 2kx)$$

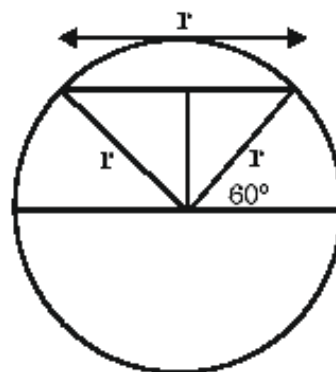
8. **Ans (A)**

Electric field



$$\frac{\rho \vec{r}_1}{3\epsilon_0} - \frac{\rho \vec{r}}{3\epsilon_0} \quad \vec{r}_1 = \vec{r} + \vec{a} \\ \vec{a} = \vec{r}_1 - \vec{r}$$

$$E_{\text{net}} = \frac{\rho \vec{r}_1}{3\epsilon_0} - \frac{\rho \vec{r}}{3\epsilon_0} \\ = \frac{\rho}{3\epsilon_0} (\vec{r}_1 - \vec{r}) = \frac{\rho}{3\epsilon_0} \vec{a}$$



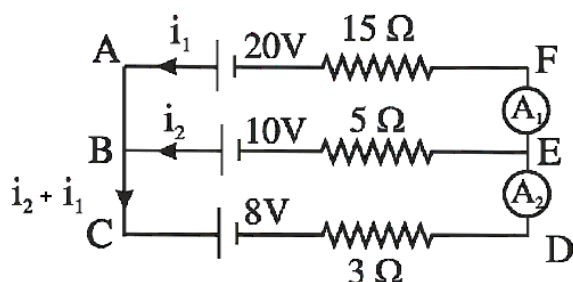
$$s = ut + \frac{1}{2} at^2$$

$$t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2r}{qE/m}}$$

$$= \sqrt{\frac{2rm \cdot 3\epsilon_0}{e \rho a}} = \sqrt{\frac{6r\epsilon_0 m}{e \rho a}}$$

9. Ans (A)

Top and middle battery polarities are interchanged.



Ammeter A_1 and A_2 are ideal, potential drop across AB and AC are zero. It means B and C are at same potential. So no current through A_3 .

KVL ABEFA

$$-10 + 5i_2 - 15i_1 + 20 = 0$$

$$3i_1 - i_2 = 2 \quad \dots (i)$$

KVL BCDEB

$$8 - 3(i_1 + i_2) - 5i_2 + 10 = 0$$

$$3i_1 + 8i_2 = 18 \quad \dots (ii)$$

$$i_1 = \frac{34}{27} \text{ A} \quad i_2 = \frac{16}{9} \text{ A}$$

$$\text{Reading of Ammeter } A_1 = i_1 + i_2 = \frac{82}{27} \text{ A}$$

$$\text{Reading of Ammeter } A_2 \quad i_1 = \frac{34}{27} \text{ A}$$

$$= i_{A_1} + i_{A_2} + i_{A_3}$$

$$= \frac{116}{27} \text{ A} = 58 \left(\frac{2}{27} \right) \text{ A}$$

10. Ans (B)

At $t = 2 \text{ sec}$

$$I = 6 \text{ A}$$

$$\frac{dI}{dt} = 2t = 2 \times 2 = 4 \text{ Amp/sec}$$

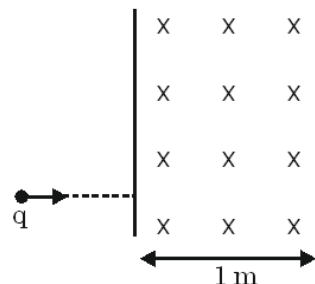
$$e = \left| \ell \frac{dI}{dt} \right| \quad e = \frac{1}{2} \times 4 = 2 \text{ V}$$

$$V_A - 2 \times 6 - e - 2 \times 6 - 3 \times 6 = V_B$$

$$V_A - V_B = 12 + 12 + 2 + 18$$

$$V_A - V_B = 44 \text{ volt}$$

11. Ans (A)



$$r = \frac{mv}{qB} = \frac{5 \times 10^{-6} \times 100}{1 \times 2.5 \times 10^{-4}} = 2 \text{ m}$$

$$\sin \theta = \frac{d}{r} = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$\theta = \pi/6$$

12. Ans (B)

$$r_{\text{mix}} = \frac{n_1 C_{p1} + n_2 C_{p2}}{n_1 C_{v1} + n_2 C_{v2}}$$

$$= \frac{4 \times \left(\frac{7}{2} R \right) + 2 \times \left(\frac{5}{2} R \right)}{4 \times \left(\frac{5}{2} R \right) + 2 \times \left(\frac{3}{2} R \right)} = \frac{19R}{13R}$$

$$\left(\frac{T'}{T} \right) = \left(\frac{V}{V} \right)^{r-1}$$

$$n = r - 1 = \frac{19}{13} - 1 = \frac{6}{13}$$

$$\text{value of } 13n = 6$$

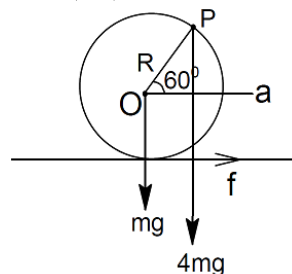
13. Ans (A)

B and C are equipotential and field is conservative

$$W_{CA} = W_{BA} = - \int_{2a}^a \frac{\lambda}{2\pi\epsilon_0 r} q dr$$

$$= \frac{\lambda q}{2\pi\epsilon_0} \ln 2$$

14. Ans (A)



$$\tau_0 = I_0 \alpha$$

$$\Rightarrow \frac{2}{5} mR\alpha + f = 2mg$$

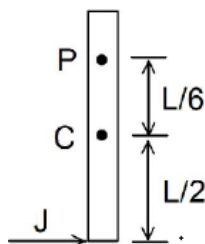
$$\text{and } f = ma = m\alpha R$$

$$\Rightarrow f = \frac{10}{7} mg \Rightarrow \frac{10}{7} mg \leq \mu N$$

$$\Rightarrow \frac{10}{7} mg \leq \mu (5mg), \mu \geq \left(\frac{2}{7} \right)$$

15. **Ans (D)**

Conservation of linear momentum.



$$J = MV_f \Rightarrow V_f = \frac{J}{M}$$

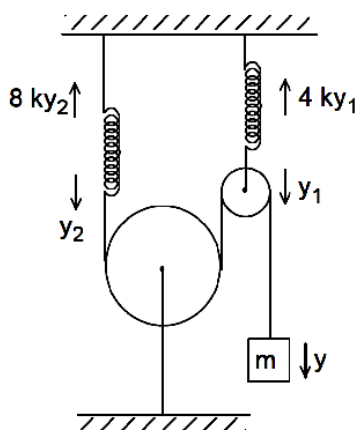
Conservation of angular momentum about C

$$J \frac{L}{2} = \frac{ML^2}{12} \omega \Rightarrow \omega = \frac{6J}{ML}$$

$$\Rightarrow KE = \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} MV_f^2 = \frac{2J^2}{M}$$

16. **Ans (B)**

$$\therefore 2F_0 = 4ky_1$$



F_0 = Increase in Tension in the string

$$F_0 = 8ky_2$$

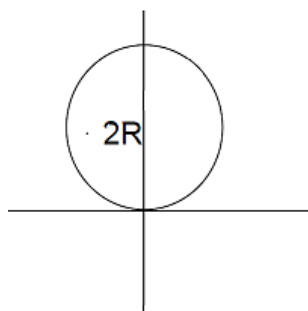
$$y = 2y_1 + y_2 = \frac{9F_0}{8K}$$

$$\Rightarrow K_{eq} = \frac{F_0}{y} = \left(\frac{8K}{9} \right)$$

$$\therefore T = 2\pi \sqrt{\frac{9m}{8K}}$$

17. **Ans (C)**

$$y = 2R = 2 \times \frac{mv_0}{qB} = \frac{2v_0}{\alpha \cdot B_0}$$



18. **Ans (B)**

The magnetic moment of straight wire,

$$m = (q_m) (\ell)$$

The magnetic moment of the semicircular wire,

$$m' = (q_m) (2r)$$

$$\therefore \frac{m'}{m} = \frac{2r}{\ell} = \frac{2r}{\pi r} = \frac{2}{\pi} \quad \therefore m' = \frac{2m}{\pi}$$

19. **Ans (B)**

The relative velocity of the diving bird relative to the fish

$$= 4 + v_B \times \frac{4}{3} = 16 \text{ cm s}^{-1}$$

$$\therefore v_B \times \frac{4}{3} = 12 \quad \therefore v_B = \frac{36}{4} = 9 \text{ cm s}^{-1}$$

20. **Ans (D)**

When the extension is maximum, their velocities are equal. From the law of conservation of Momentum,

$$P_f = p_i \Rightarrow (6)v + (3)v = 6(2) + 3(-1)$$

$$v = 1 \text{ m s}^{-1}$$

Energy is also conserved

$$E_f = E_i \Rightarrow \frac{1}{2} (6) (1)^2 + \frac{1}{2} (3) (1)^2 + \frac{1}{2} Kx_m^2$$

$$= \frac{1}{2} (6) (2)^2 + \frac{1}{2} (3) (1)^2$$

$$3 + 1.5 + \frac{1}{2} (200) x_m^2 = 12 + 1.5$$

$$100x_m^2 = 9 \Rightarrow x_m^2 = 0.09$$

$$\Rightarrow x_m = 0.3 \text{ m} = 30 \text{ cm}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (10)**

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0} \left(\frac{1}{x} \hat{i} + \frac{1}{y} \hat{j} \right)$$

$$\text{Field lines : } \frac{dy}{dx} = \frac{E_y}{E_x} = \frac{x}{y}$$

$$\text{Equipotential : } \frac{dy}{dx} = \frac{-y}{x}$$

$$\therefore xy = 50$$

$$\text{Closest point} = (5\sqrt{2}, 5\sqrt{2})$$

$$\text{Nearest distance} = 5\sqrt{2} \times \sqrt{2}$$

$$= 10 \text{ m}$$

2. **Ans (27)**

Initially, $I = 40/10 = 4$ A. The total flux remains constant. Hence the new current is given by $5 \times 4 = 15 \times I'$ and hence

$$I' = 4/3 \text{ A.}$$

Energy lost

$$= 0.5 \times 5 \times 4^2 - 0.5 \times 15 \times (4/3)^2$$

$$= 27 \text{ Joules}$$

3. **Ans (40)**

The distance covered by particle

$$= 6 \times 5 = 30 \text{ m.}$$

$$5 \cos \theta = 3$$

$$\text{Hence } \cos \theta = 0.6$$

Hence distance of closest approach

$$= 30 \tan \theta = 40 \text{ m.}$$

4. **Ans (13)**

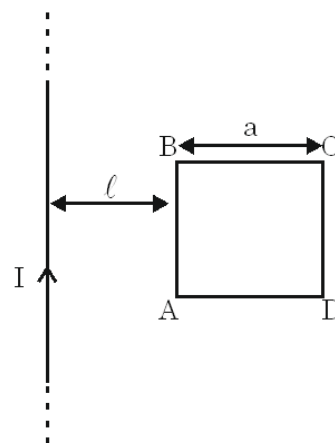
$$\begin{aligned} KE &= \frac{1}{2} m' v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2 \\ &= \frac{1}{2} \times \frac{m}{2} \times \left(v - \frac{v}{R} \times \frac{3R}{8} \right)^2 \\ &+ \frac{1}{2} \left(\frac{mR^2}{2} \times \frac{2}{5} - \frac{m}{2} \left(\frac{3R}{8} \right)^2 \right) \times \left(\frac{v}{R} \right)^2 \\ &= \frac{13}{80} mv^2 = \frac{13}{80} \times 5 \times 4^2 = 13 \text{ Joules} \end{aligned}$$

5. **Ans (375)**

$$\begin{aligned} A &= \frac{900\pi}{9 \times 10^8} = 10^{-6} \pi \text{ m}^2 \\ L &= \frac{m}{PA} = \frac{4\pi \times 10^{-3}}{10^4 \times 10^{-6} \pi} = \frac{4 \times 10^3}{10^4} = 0.4 \text{ m} \\ v &= \frac{1}{2L} \sqrt{\frac{T}{\mu}} = \frac{1}{2 \times 0.4} \sqrt{\frac{900\pi}{4\pi \times 10^{-3}/L}} \\ &= \frac{1}{2\sqrt{0.4}} \times \sqrt{225 \times 10^3} \\ &= \frac{1}{2\sqrt{0.4}} \times 15 \times \sqrt{10^3} \\ &= \frac{15}{2} \times 50 = 15 \times 25 = 375 \text{ Hz} \end{aligned}$$

6. **Ans (3)**

General



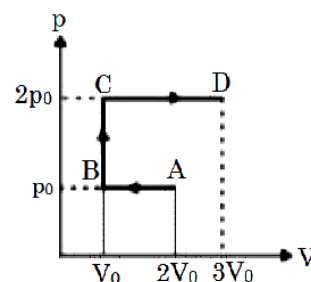
$$\phi_{ABCD} = \frac{\mu_0 I a}{2\pi} \ln \left(\frac{a+l}{a} \right)$$

$$\text{Also, } q = \frac{\Delta \phi}{R}$$

q : charge flown

R : Resistance of loop

7. **Ans (9)**



Monoatomic gas

$$C_p = \frac{5}{2} R; \quad C_v = \frac{3}{2} R$$

$$\begin{aligned} Q_{AB} &= nC_p(T_B - T_A) \\ &= n \left(\frac{5}{2} R \right) \left(\frac{p_0 V_0}{nR} - \frac{p_0 (2V_0)}{nR} \right) = -\frac{5}{2} p_0 V_0 \end{aligned}$$

$$\begin{aligned} Q_{BC} &= nC_v(T_C - T_B) \\ &= n \left(\frac{3}{2} R \right) \left(\frac{2p_0 V_0}{nR} - \frac{p_0 V_0}{nR} \right) = \frac{3}{2} p_0 V_0 \end{aligned}$$

$$\begin{aligned} Q_{CD} &= nC_p(T_D - T_C) \\ &= n \left(\frac{5}{2} R \right) \left(\frac{2p_0 (3V_0)}{nR} - \frac{2p_0 (V_0)}{nR} \right) = 10p_0 V_0 \end{aligned}$$

$$\begin{aligned} Q_{\text{Total}} &= nC(T_D - T_A) \\ &= 10p_0 V_0 - \frac{5}{2} p_0 V_0 + \frac{3}{2} p_0 V_0 \\ nC \left(\frac{6p_0 V_0}{nR} - \frac{2p_0 V_0}{nR} \right) &= 9p_0 V_0 \end{aligned}$$

$$C = \frac{9R}{4} = \frac{xR}{4}$$

$$x = 9$$

8. Ans (2)

$$W_{\text{all}} = \Delta KE$$

$$\int \vec{F} \cdot d\vec{r} = \Delta KE$$

$$\int \left((-2x + 2y)\hat{i} + (2x - y^2)\hat{j} \right) \cdot (dx\hat{i} + dy\hat{j}) = \Delta KE$$

$$\int -2x dx + \int 2y dx + \int 2x dy - \int y^2 dy = \Delta KE$$

$$\int_1^2 -2x dx + \int_3^2 2y dy + \int_1^2 2x dx - \int_3^2 y^2 dy$$

$$= K_{(1,2)} - K_{(2,3)}$$

$$-x^2 \Big|_1^2 + y^2 \Big|_3^2 + x^2 \Big|_1^2 - \frac{y^3}{3} \Big|_3^2 = K_{(1,2)} - \frac{8}{3}$$

$$-9 + 4 + \left(+9 - \frac{8}{3} \right)$$

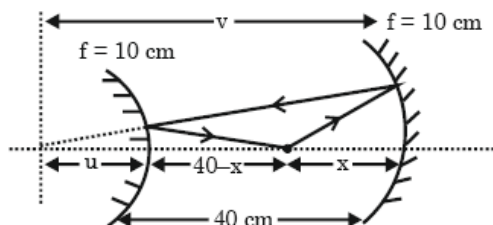
$$-5 + \frac{19}{3} = KE_{(1,2)} - \frac{8}{3}$$

$$KE_{1,2} = 4J$$

$$\frac{1}{2}mv^2 = 4$$

$$\frac{1}{2} \times 2v^2 = 4 \Rightarrow \boxed{v = 2 \text{ m/s}}$$

9. Ans (13)



From Concave Mirror :

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{x} = \frac{-1}{10}$$

$$\Rightarrow \boxed{v = \frac{10x}{10 - x}} \dots\dots(1)$$

From Convex Mirror :

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{-(40 - x)} + \frac{1}{u} = \frac{1}{10}$$

$$\Rightarrow \boxed{u = \frac{10(x - 40)}{(x - 50)}} \dots\dots(2)$$

From figure its clear that

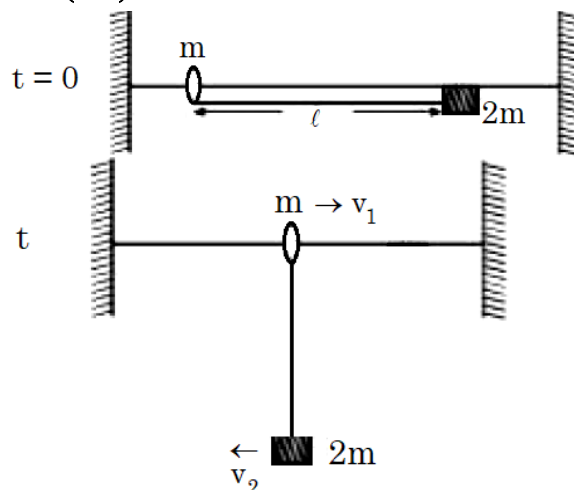
$$v = u + 40$$

$$\frac{10(x)}{10 - x} = \frac{10(x - 40)}{x - 50} + 40$$

On solving

$$x \approx 13 \text{ cm}$$

10. Ans (70)



$$1. \quad mv_1 - 2mv_2 = 0 \quad \{p_i = p_f\}$$

$$v_1 = 2v_2 \quad \dots\dots(i)$$

$$2. \quad W_g = \Delta KE$$

$$2mg\ell = \frac{1}{2}mv_1^2 + \frac{1}{2}2mv_2^2 \quad \dots\dots(ii)$$

Solving equation (i) and (ii)

$$v_2 = \sqrt{\frac{2g\ell}{3}} \quad v_1 = \sqrt{\frac{8g\ell}{3}}$$

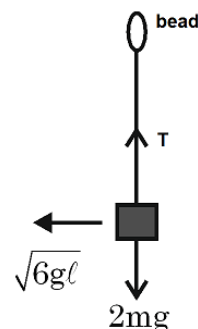
$$|\vec{v}_{2,1}| = |\vec{v}_2 - \vec{v}_1| = \sqrt{6g\ell}$$

from bead frame

$$T - 2mg = \frac{2m(\sqrt{6g\ell})^2}{\ell}$$

$$T = 14mg$$

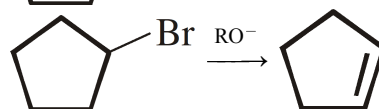
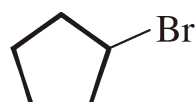
$$T = 14 \left(\frac{1}{2} \right) \times 10 = 70 \text{ N}$$



PART-2 : CHEMISTRY

SECTION-I

1. Ans (B)



only one E2 product

3. Ans (C)

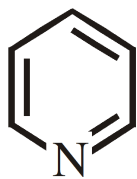
Volume was increased

4. Ans (D)

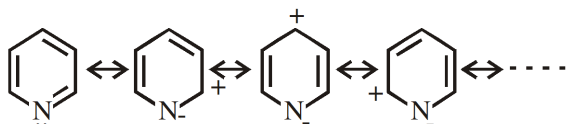
H; HH; HH

More substituted alkene is more stable (thermodynamically).

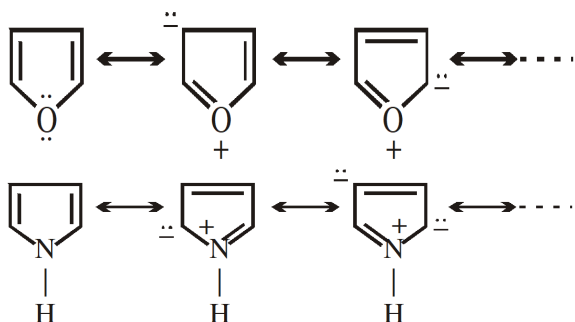
5. **Ans (A)**



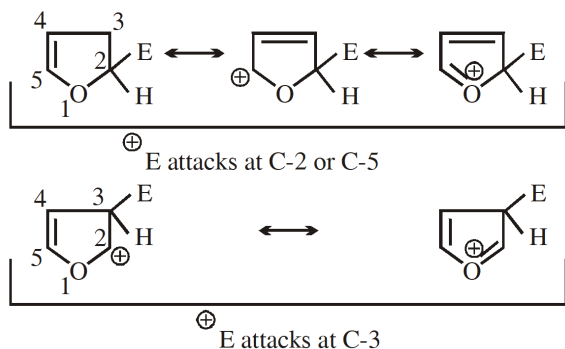
Pyridine is least reactive towards electrophilic substitution because pyridine is resonance stabilised as shown



Because of withdrawal of electrons from the ring by N atom, the ring is deactivated, So, it is least reactive towards SE reaction.



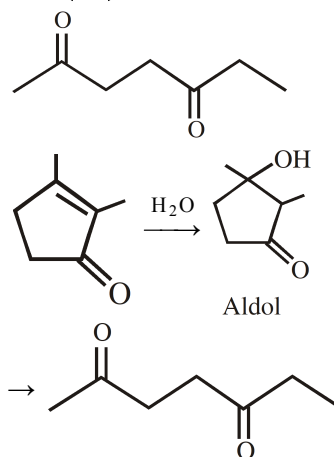
SE reaction in pyrrole, thiophene, and furan takes place at C-2 or C-5 positions because there are three resonating structures when E^+ attacks at C-2 or C-5. and two resonating structures when E^+ attacks at C-3.



6. **Ans (B)**

Heating metallic iron in a stream of dry chlorine gas.

7. **Ans (C)**



8. **Ans (A)**

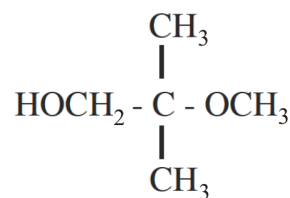
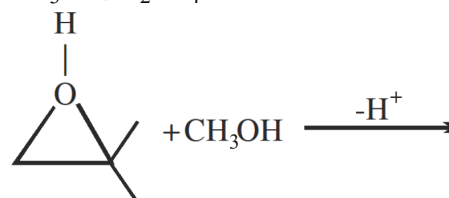
$$t_{1/2} \propto \frac{1}{[A]^{n-1}}$$

$$\Rightarrow \frac{(t_{1/2})_2}{(t_{1/2})_1} = 8 = \left(\frac{[A]_{01}}{[A]_{02}} \right)^{n-1} = 2^{n-1}$$

9. **Ans (D)**

presence of many hydroxyl groups on the polymer backbone. Cellulose is a polyhydroxy compound, contain several free hydroxy group on its backbone capable of forming Intermolecular H-bonds with water.

10. **Ans (B)**



11. **Ans (C)**

$$2(\sqrt{2})^3 : (\sqrt{3})^3$$

$$a_{fcc} = 2\sqrt{2} r$$

$$a_{bcc} = \frac{4r}{\sqrt{3}}$$

$$\text{density} = \frac{Z \times M}{N_A \times a^3}$$

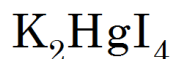
12. **Ans (A)**

The colloidal particles migrate in an applied electric field. When electric potential is applied across two Pt-electrodes, dipping in a colloidal solution the particles move towards positive or negative ends or electrodes.

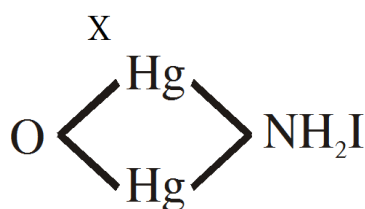
13. **Ans (D)**

Desilverisation process

14. **Ans (A)**



(Nessler's reagent)

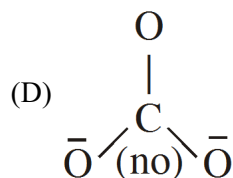
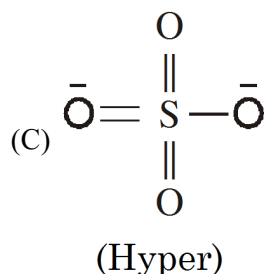
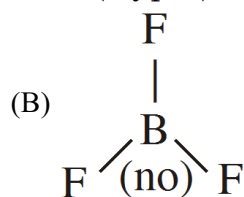
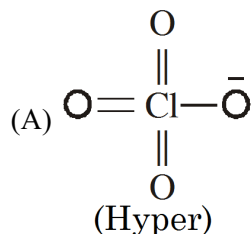


(Iodide of Million's base)

Y

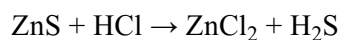
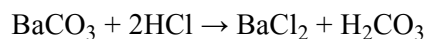
15. **Ans (B)**

Hypervalent $\Rightarrow$ violating octet rule



16. **Ans (B)**

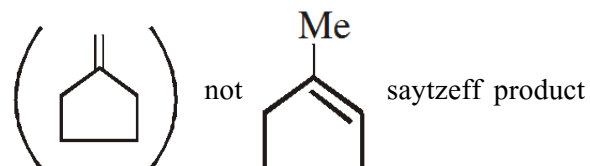
BaCO_3 and ZnS insoluble in water but soluble in dilute HCl solution



FeCl_3 and CaCO_3 also soluble on addition of HCl but produces yellow solution

17. **Ans (C)**

Reagent (C) is a bulky base, it will give



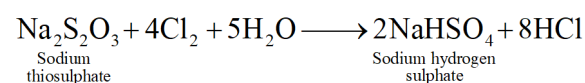
obtained by reagent a, b, d (less-bulky base)

18. **Ans (C)**

SO_3 acts as an electrophile in electrophile substitution reaction.

19. **Ans (B)**

When aqueous sodium thiosulphate reacts with Cl_2 then form sodium hydrogen sulphate.



20. **Ans (A)**

Enantiomers have same magnitude of optical rotations but opposite sign

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (21)**

Given, wavelength (λ) = 10 nm

Form λ and KE relationship, $\lambda = \frac{h}{\sqrt{2KEm}}$

where, h = planks constant = 6.62×10^{-34} Js

Mass of electron (m) = 9.1×10^{-31} kg

$$\text{Thus, } 10 \times 10^{-9} = \frac{6.62 \times 10^{-34} \text{ Js}}{\sqrt{2 \times KE \times 9.1 \times 10^{-31} \text{ kg}}}$$

$$\sqrt{2KE} = \frac{6.62 \times 10^{-34}}{\sqrt{9.1 \times 10^{-31} \times 10^{-9}}}$$

$$KE = 2.4 \times 10^{-21} \text{ J}$$

2. **Ans (47)**

$$E^\circ = 0.34 - (-0.13) = 0.47 \text{ V}$$

3. **Ans (139)**

For step II : $p^{-\frac{1-\gamma}{\gamma}} T = \text{constant}$

$$p_2^{-\frac{1-\gamma}{\gamma}} T_2 = p_3^{-\frac{1-\gamma}{\gamma}} T_3$$

$$p_3 \left(\frac{T_2}{T_3} \right)^{\frac{\gamma}{1-\gamma}} p_2 = \left(\frac{600}{300} \right)^{-\frac{7}{2}} \times 2$$

$$= \frac{1}{2^{7/2}} \times 2 = \frac{1}{4\sqrt{2}} \text{ atm}$$

$$V_3 = \frac{RT_3}{p_3} = \frac{0.082 \times 300}{\frac{1}{4\sqrt{2}}} = 139 \text{ L}$$

4. **Ans (67)**

Vapour pressure of two isomeric liquid X and Y are 200 torr and 100 torr respectively. So, total pressure of two liquid X, Y

$$p_T = p_x + p_y$$

$$= \frac{200}{2} + \frac{100}{2} = 100 + 50 = 150 \text{ torr}$$

From partial pressure equation,

$$\text{For X liquid } (p_T)y_1 = 100 \text{ torr. (i)}$$

$$\text{For Y liquid } (p_T)y_2 = 50 \text{ torr. (ii)}$$

$$\text{From eq. (i) } \div \text{ (ii), } \frac{y_1}{y_2} = \frac{100/150}{50/150} = \frac{2}{1}$$

So, mole fraction of X = $2/3 = 0.667$

5. **Ans (129)**

Given conductivity (k) = $1.29 \times 10^{-2} \text{ Scm}^{-1}$

Resistance (R) = 28.44Ω

From equation of molar conductivity,

$$\lambda_m = \frac{k \times 1000}{\text{molarity/concentration (mol L}^{-1}\text{)}}$$

$$\text{or } \lambda_m = \frac{\text{Conductance (c) } \times \text{ cell constant (G*) } \times 1000}{\text{molarity/concentration (m)}}$$

$$\text{or } \lambda_m = \frac{\left[\frac{1}{R} \times \frac{1}{a} \right] \times 1000}{(m)} \quad \dots(i)$$

For calculation of cell constant

$$k = \frac{1}{R} \times \frac{1}{a}; 1.29 \times 10^{-2} \text{ Scm}^{-1} = \frac{1}{28.44\Omega} \times \frac{1}{a}$$

$$\text{or } \frac{1}{a} = 1.29 \times 10^{-2} \times 28.22 \quad \dots(ii)$$

Now, put the value from eq (ii) in eq (i)

$$\lambda_m = \frac{1}{28.44} \times \frac{1.29 \times 10^{-2} \times 28.44 \times 1000}{0.5}$$

$$= 130 \text{ Scm}^2 \text{ mol}^{-1}$$

6. **Ans (12)**

$$K_C = \frac{x}{2(2)^2} = \frac{1}{2}$$

$x = 4 \text{ mol AB}_2 \text{ present.}$

Now let mol of AB_2 is added

$$K_C = \frac{1}{2} = \frac{4+y}{2(2+2)^2} \Rightarrow y = 12$$

7. **Ans (40)**

$$A_0(X) = 8 A_0(Y)$$

$$A_t(X) = A_t(Y)$$

$$\Rightarrow 8e^{\frac{-\ln 2}{20} \times 120} = e^{\frac{-\ln 2}{T} \times 120}$$

Solving

$T = \text{Half-life}$

$= 40 \text{ min}$

8. **Ans (3)**

$$(\Delta p)_{\text{glucose}} = (\Delta p)_{\text{urea}}$$

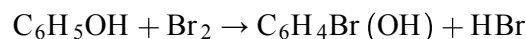
$$(x_B)_{\text{glucose}} = (x_B)_{\text{urea}}$$

$$\text{i.e., } \left(\frac{n_B}{n_A} \right)_{\text{glucose}} = \left(\frac{n_B}{n_A} \right)_{\text{urea}}$$

$$\frac{w_B}{50} \times \frac{18}{180} = \frac{1 \times 18}{50 \times 60}$$

$$w_B = 3 \text{ g}$$

9. **Ans (14)**



Moles of Br_2 required

$$= \frac{500 \times 1000}{94} = 5319.15 \text{ mole}$$

$$\Rightarrow \text{g equivalent of } \text{Br}_2 = 5319.15 \times 2$$

$$\Rightarrow t = \frac{Q}{I} = \frac{5319.15 \times 2 \times 96500}{20000} = 51329.8 \text{ s}$$

$$= 14.26 \text{ hour}$$

10. **Ans (6)**

$$\Delta H = \Delta U + \Delta (pV)$$

$$= 14 + (2 - 10) = 4 \text{ L atm.}$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (D)**

$$f(2) = 0, f(2-0) = 1 - 1 = 0,$$

$$f(2+0) = 0 - 0 = 0$$

$$f(-2) = 0, f(-2-0) = 0 - 1 = -1$$

$$\Rightarrow f(x) \text{ is continuous at } x = 2 \text{ but not at } x = -2$$

2. **Ans (C)**

$$(z+1)^7 + z^7 = 0$$

$$\Rightarrow (z+1)^7 = -z^7$$

$$\Rightarrow |z+1| = |z|$$

$$\Rightarrow z \text{ lies on the line } x = -\frac{1}{2}$$

$$\therefore \sum_{k=0}^6 \operatorname{Re}(z_k) = -\frac{7}{2}$$

3. **Ans (B)**

$$\text{Let } I = \int_0^1 f(x) dx$$

$$I = \int_0^{1/2} f(x) dx + \int_0^{1/2} f(1-x) dx$$

$$= \int_0^{1/2} f(x) dx + \int_0^{1/2} f\left(1 - \left(\frac{1}{2} - x\right)\right) dx$$

$$= \int_0^{1/2} f(x) dx + \int_0^{1/2} f\left(\frac{1}{2} + x\right) dx$$

$$= \int_0^{1/2} \left(f(x) + f\left(\frac{1}{2} + x\right)\right) dx$$

$$= \int_0^{1/2} 1 dx = \frac{1}{2} = k.$$

$$\text{Hence } 2k = 1$$

4. **Ans (C)**

$$\text{Hence } \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

$$\Rightarrow (a+b+c)(a^2+b^2+c^2-ab-bc-ca) = 0$$

$$\Rightarrow a+b+c = 0 \quad (\because a \neq b \neq c)$$

$$\therefore \text{point of intersection is } (1, 1)$$

$$\therefore 3+2-\lambda = 0 \Rightarrow \lambda = 5$$

5. **Ans (C)**

$$f(x) = 64x^3 + \frac{1}{x^3} = \left(4x + \frac{1}{x}\right)^3 - 3.4\left(4x + \frac{1}{x}\right)$$

$$\text{So } f(\alpha) = 2^3 - 3.4.2 = -16$$

$$\text{Also } f(\beta) = 2^3 - 3.4.2 = -16$$

6. **Ans (B)**

$$\sim (\sim p \wedge q) \wedge (p \vee q) = [\sim (\sim p) \vee (\sim q)] \wedge (p \vee q)$$

$$= [p \vee (\sim q) \wedge (p \vee q)]$$

$$= p \vee [(\sim q) \wedge q]$$

$$= p \vee f = p$$

7. **Ans (D)**

$$f(x) = \cos^{-1} \left(\frac{\sqrt{2x^2+1}}{x^2+1} \right)$$

$$f(x) = \sin^{-1} \left(\frac{x^2}{x^2+1} \right)$$

$$0 \leq \frac{x^2}{x^2+1} < 1$$

$$0 \leq \sin^{-1} \left(\frac{x^2}{x^2+1} \right) < \frac{\pi}{2}$$

$$0 \leq f(x) < \frac{\pi}{2}$$

$$\text{Range } \left[0, \frac{\pi}{2}\right)$$

8.

Ans (B)

$$\lim_{\theta \rightarrow 0} \left(\frac{1+a \cos \theta}{\theta^2} - \frac{b \sin \theta}{\theta^3} \right) = 1$$

$$\lim_{\theta \rightarrow 0} \frac{\left(\theta \left[1+a \left(1-\frac{\theta^2}{2}\right)\right] - b \left[\theta - \frac{\theta^3}{3!}\right]\right)}{\theta^3} = 1$$

$$\Rightarrow \lim_{\theta \rightarrow 0} \left(\frac{\theta(1+a-b) - \frac{a\theta^3}{2} + \frac{b\theta^3}{6}}{\theta^3} \right) = 1$$

$$\therefore 1+a-b=0$$

$$\text{and } \frac{b}{6} - \frac{a}{2} = 1$$

$$\Rightarrow a = -5/2$$

$$b = -3/2$$

$$\Rightarrow b+a = -4$$

$$\therefore \lambda = 4$$

9. **Ans (A)**

$$\Delta = \begin{vmatrix} \lambda & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = 0$$

$$C_1 \rightarrow C_1 + C_2 + C_3$$

$$(\lambda + 2) \begin{vmatrix} 1 & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = 0$$

$$C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$$

$$\Rightarrow (\lambda + 2)(\lambda - 1)^2 = 0$$

10. **Ans (B)**

From the formula of mean we have

$$14.7 = \frac{114 + a + b}{10} \Rightarrow a + b = 33$$

from the formula of variance we have

$$\sigma^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n}$$

$$10.01 = \frac{69.12 + (a - 14.7)^2 + (b - 14.7)^2}{10}$$

$$\text{put } a = 33 - b$$

$$100.1 = 2b^2 - 66b + 620.1$$

$$\Rightarrow b^2 - 33b + 260 = 0$$

$$\Rightarrow b = 13 \text{ or } 20$$

$$\text{So } a = 20 \text{ or } 13$$

$$2a + b \text{ is either } 46 \text{ or } 53$$

11. **Ans (A)**

$$S_n = \frac{\left(1 - \left(\frac{1}{2}\right)^n\right)}{1 - \frac{1}{2}}$$

$$S_n = 2 - \left(\frac{1}{2}\right)^{n-1} \Rightarrow \left(\frac{1}{2}\right)^{n-1} < \frac{1}{100}$$

$$\Rightarrow n - 1 \geq 7 \Rightarrow n \geq 8$$

12. **Ans (D)**

$$A = \frac{\pi}{2} \Rightarrow b^2 + c^2 = a^2$$

$$(b + c)^2 - 2bc = a^2$$

$$(b + c)^2 - a^2 = 2bc$$

$$= 4\Delta$$

$$(b + c - a)(b + c + a) = 4(a + b + c)$$

$$\Rightarrow b + c - a = 4$$

13. **Ans (A)**

$$\frac{f(3) - f(1)}{3 - 1} = f'(c), \text{ for some } c \text{ such that}$$

$$1 < c < 3$$

$$f'(c) = \frac{f(3) - 2}{2} < 2$$

$$\therefore f(3) < 6$$

14. **Ans (C)**

$$\sqrt{1 + \sec x} = \sqrt{\frac{1 + \cos x}{\cos x}} = \frac{\sqrt{2} \cos \frac{x}{2}}{\sqrt{1 - 2\sin^2 \frac{x}{2}}}$$

$$\text{Put } \sin \frac{x}{2} = t$$

$$\int \sqrt{1 + \sec x} dx = \int \frac{\sqrt{2} \cos \frac{x}{2}}{\sqrt{1 - 2\sin^2 \frac{x}{2}}} dx$$

$$= 2\sin^{-1} \sqrt{2}t + C$$

$$= 2\sin^{-1} \left(\sqrt{2} \sin \frac{x}{2} \right) + C$$

15. **Ans (D)**

$$\frac{dy}{dx} = \frac{e^{-3x}}{-3} + c_1$$

$$y = \frac{e^{-3x}}{9} + c_1x + c_2$$

16. **Ans (B)**

If A is an involutory and diagonal matrix, then

$$A^2 = I.$$

$$\text{Let } A = \begin{vmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{vmatrix}. \text{ Then } A^2 = I$$

$$\Rightarrow a^2 = 1$$

$$\Rightarrow a = -1 \quad \{\because a \text{ is non-positive}\}$$

$$\text{Thus } |A| = -1$$

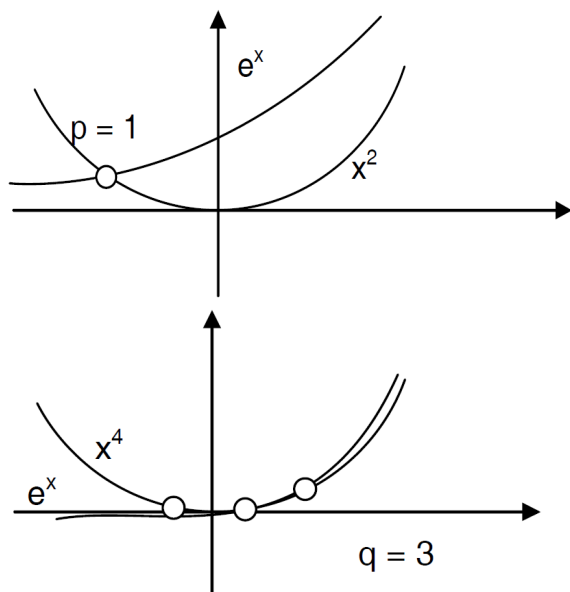
$$A^{-1} = A$$

$$|2A| = -8$$

$$|3 \text{ adj } (2A)| = 27 \cdot |\text{adj } (2A)|$$

$$= 27(-8)^2 = 1728$$

17. Ans (D)



$$p + q = 4$$

18. Ans (A)

$$\begin{aligned} \text{Ordered pair} &= \text{total} - (A \cup B = X) \\ &= 4^n - 3^n \end{aligned}$$

19. Ans (C)

$$\begin{aligned} \text{Given } f &= \{(1, 3), (2, 4), (3, 2), (4, 1)\} \\ \therefore f \circ f &= \{(1, 2), (2, 1), (3, 4), (4, 3)\} \\ \therefore f \circ f \circ f &= \{(1, 1), (2, 2), (3, 3), (4, 4)\} \\ \underbrace{f \circ f \circ f \dots \text{of}(x)}_{2012 \text{ times}} &= \{(1, 1), (2, 2), (3, 3), (4, 4)\} \end{aligned}$$

20. Ans (A)

$$\text{We have } \vec{V}_1 = \lambda \vec{V}_2 \text{ (given)}$$

$$\text{So, } \frac{3ax^2}{b(x-1)} = \frac{-2(x-1)}{x^2}$$

$$\Rightarrow 3ax^4 + 2b(x-1)^2 = 0$$

$$f(x) = 3ax^4 + 2b(x-1)^2$$

$$f(0) = 2b$$

$$f(1) = 3a$$

$$\Rightarrow f(0) f(1) < 0$$

$$\Rightarrow f(0) \text{ and } f(1) \text{ have opposite signs.}$$

$$\text{Hence } f(x) \text{ must have a root in } (0, 1).$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (7)

$$2 = a + b + \frac{7}{2} \quad \dots(i)$$

$$\frac{dy}{dx} = 2x + 6 \text{ at } (-2, 2)$$

$$\frac{dy}{dx} = 2ax + b \text{ at } (1, 2)$$

$$\Rightarrow m = 2a + b$$

$$2a + b = -\frac{1}{2} \quad \dots(ii)$$

$$\text{from (i) and (ii) } a = 1, b = -5/2$$

$$2(a - b) = 7$$

2. Ans (8)

$$\text{coefficient of } t^8 \text{ in expansion of}$$

$$(1 + t^2)^{12} (1 + t^{12} + t^{24} + t^{36})$$

$$= \text{coefficient of } t^8 \text{ in expansion of } (1 + t^2)^{12}$$

$$= {}^{12}C_4$$

$$\Rightarrow m = 12 \text{ and } n = 4$$

$$\Rightarrow m - n = 8$$

3. Ans (13)

$$\text{Equation of such circle is}$$

$$x^2 + y^2 - \frac{10}{3}x - 2y - 15 = 0$$

$$\text{so radius } r = \sqrt{\frac{25}{9} + 1 + 15} = \sqrt{\frac{169}{9}} = \frac{13}{3}$$

$$\Rightarrow 3r = 13$$

4. Ans (80)

$$\begin{aligned} \text{Select 4 pairs in } {}^5C_4 = 5 \text{ ways. Now select} \\ \text{exactly one shoe from each of the selected} \\ \text{pair} = {}^2C_1 \times {}^2C_1 \times {}^2C_1 \times {}^2C_1 = 16 \text{ ways. This} \\ \text{will fulfill the condition.} \end{aligned}$$

$$\text{Required answer} = 5 \times 16 = 80$$

5. Ans (1)

$$y^2 + 2y + 1 = -x - 1$$

$$(y + 1)^2 = -(x + 1)$$

$$\text{Equation of normal will be}$$

$$y + 1 = m(x + 1) + \frac{m}{2} + \frac{1}{4}m^3$$

$$\text{Since, it passes through } (2, -1) :$$

$$0 = 3m + \frac{m}{2} + \frac{m^3}{4}$$

$$\Rightarrow m = 0 \text{ or } m^2 + 14 = 0 \text{ (not possible)}$$

6. **Ans (4)**

5	4	4	4	4	4	4	4	4	4	4

Favourable case = 5×4^9

Total case = 5^{10}

$$\text{Probability} = \frac{5 \cdot 4^9}{5^{10}} = \left(\frac{4}{5}\right)^9$$

7. **Ans (9)**

A(-3, 6, 0) is a point on first line

B(-2, 0, 7) is a point on second line

direction ratios of the line of shortest distances are

$$\frac{a}{3-2} = \frac{b}{-8+4} = \frac{c}{-4+12}$$

i.e. $\langle 1, -4, 8 \rangle$

$\therefore$ shortest distance

$$= \frac{(-2+3) \cdot 1 + (0-6)(-4) + (7-0) \cdot 8}{\sqrt{1+16+64}}$$

$$= \frac{81}{9} = 9$$

8. **Ans (2)**

$$\alpha + \beta = \frac{\pi}{2} \Rightarrow \tan \alpha \tan \beta = 1$$

$$\gamma = \alpha - \beta$$

$$\tan \gamma = \tan (\alpha - \beta)$$

$$\tan \gamma = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\tan \gamma = \frac{\tan \alpha - \tan \beta}{2}$$

$$\Rightarrow \frac{\tan \alpha - \tan \beta}{\tan \gamma} = 2$$

9. **Ans (0)**

$$\lim_{h \rightarrow 0} \frac{f(3+h^2) + f(3-h^2) - 2f(3)}{2h^2}$$

$$= \lim_{h \rightarrow 0} \frac{1}{2} \left(\frac{f(3+h^2) - f(3)}{h^2} - \frac{f(3-h^2) - f(3)}{-h^2} \right)$$

$$= \frac{1}{2} (f'(3) - f'(3)) = 0$$

10. **Ans (0)**

$$x + y = 60$$

$$y = 60 - x$$

$$x^3 y = (60 - x)(x^3)$$

$$f(x) = (60 - x)x^3, x \in (0, 60)$$

$$f'(x) = 3x^2(60 - x) - x^3 = 0$$

$$\text{or } x = 45 \quad (\because x \neq 0)$$

$$f'(45^+) < 0 \text{ and } f'(45^-) > 0$$

Hence maximum at $x = 45$

$$\Rightarrow x = 45, y = 15$$